

HAT-P-36b

R-1920

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-36_210601 · created 2026-08-06T00:43:38+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459366.7101 +/- 0.0041 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.077 +/- 0.028
Transit depth (Rp/Rs)^2	0.59 +/- 0.44 [%]
Orbital Inclination (inc)	81.91 +/- 2.79
Airmass coefficient 1 (a1)	0.758 +/- 0.012
Airmass coefficient 2 (a2)	0.221 +/- 0.026
Scatter in the residuals of the lightcurve fit is	1.59 %
Best Comparison Star	#2 - [265, 355]
Optimal Aperture	3.82
Optimal Annulus	10.85
Transit Duration (day)	0.072 +/- 0.017

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.077 $\pm$ 0.028 vs 0.126 (deviation 1.55 σ)
Duration fit vs published	0.072 d vs 0.0925 d (deviation 1.07 σ)
Mid-time O-C	10.0 min
Depth SNR	2.4
Residual scatter	1.59 %

Light curve

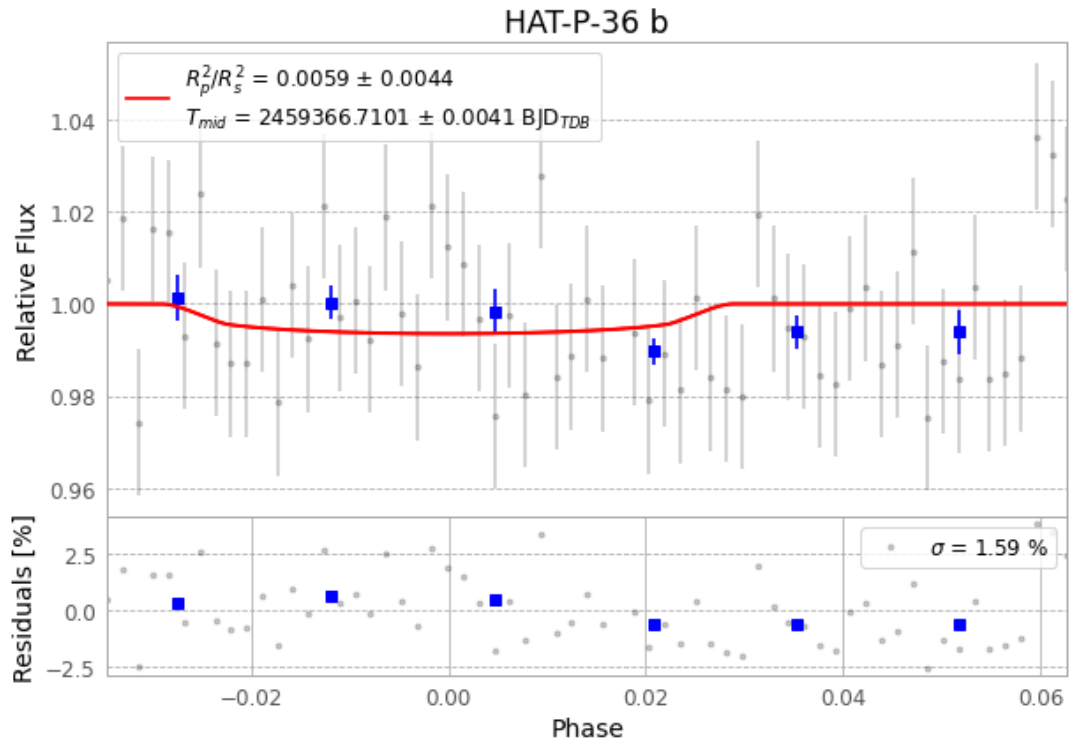


Figure 1 — FinalLightCurve_HAT-P-36 b_2021-05-31.png (SHA-256 4f098820f795286c...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [409, 210], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 63bc754d2d19f002...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

63 FITS frames (41 MB), HATP-36210601035412.FITS → HATP-36210601070012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-36-210601.log (34dbed3315b9...), FinalLightCurve_HAT-P-36 b_2021-05-31.png (4f098820f795...), FinalLightCurve_HAT-P-36 b_2021-05-31.csv (3d7fe7475599...)

Compute resources

Wall time 192.9 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-06 00:43Z.